Chapter 9 Notes

Section 1

Notes

Here is why $A \cup B = A \cup (B - A)$.

On the one hand, suppose that $x \in A \cup B$. Then either $x \in A$ or $x \in B$. In the first case, obviously $x \in A \cup (B - A)$. In the second case, if $x \in A$, then we are done (by the first case). Otherwise $x \in B - A$, meaning $x \in A \cup (B - A)$. On the other hand, suppose that $x \in A \cup (B - A)$. Then obviously $x \in A \cup B$.

Section 2

Notes

Note that Lemma 2.2 assumes that h is defined on all subsets of A.

In proof 2 of the Bernstein-Schröder theorem, here is why $A - [A - g''(B - f''(X))] = g''(B - f''(X))$. Obviously, if $x \in A - [A - g''(B - f''(X))]$, then $x \in g''(B - f''(X))$. On the other hand, if $x \in g''(B - f''(x))$, note that $x \in A$, as g is a function from B into A. Thus $x \notin A - g''(B - f''(X))$, meaning $x \in A - [A - g''(B - f''(X))]$.

Exercise 2.1

(a) Consider any fixed point A of h. Then $h(A) = A$, meaning $A \subseteq h(A)$. Hence $A \in \Sigma$, and thus $A \subseteq B$. Thus B is the largest fixed point of h.

(b) Let Σ be the set of all subsets X of A such that $X \subseteq h(X)$ and that X is a subset of all fixed points of h. Let $B = \cup\Sigma$. We aim to show that B is a fixed point of h. Obviously, B is a subset of all fixed points of h (thus B is the smallest fixed point of h).

On the one hand, we show that $B \subseteq h(B)$ (this part is identical to the proof of the first part of the original lemma). So suppose that $b \in B$. Then $b \in X$ for some $X \in \Sigma$. Because $X \in \Sigma$, then $X \subseteq h(X)$, hence $b \in h(X)$. But $X \subseteq B$, and because h is monotone, it follows that $h(X) \subseteq h(B)$. Thus $x \in h(B)$. Hence $B \subseteq h(B)$.

On the other hand, we show that $h(B) \subseteq B$. Well, we have shown that $B \subseteq h(B)$, meaning $h(B) \subseteq h(h(B))$ (because h is monotone). Also, consider any fixed point A of h. Then $B \subseteq A$. Hence $h(B) \subseteq h(A)$. But $h(A) = A$, meaning $h(B) \subseteq A$. Thus $h(B)$ is a subset of every fixed point of h, and $h(B) \subseteq h(h(B))$, meaning $h(B) \in \Sigma$, and thus $h(B) \subseteq B$. Hence we have shown that $h(B) = B$, meaning B is a fixed point of h (and the smallest one).